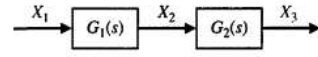
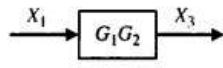
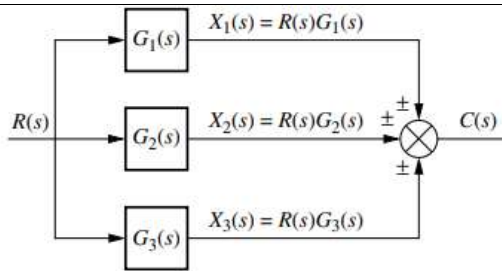
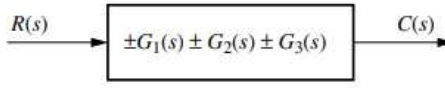
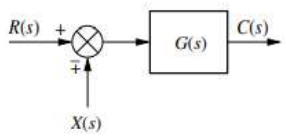
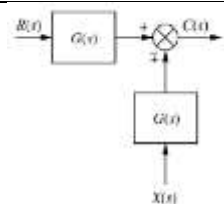
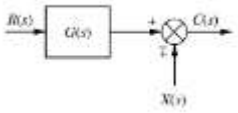
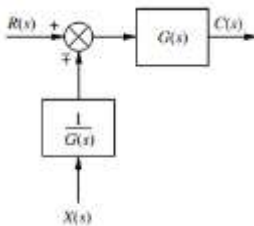
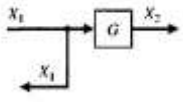
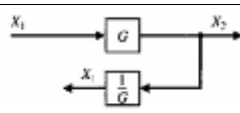
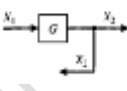
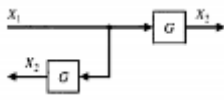
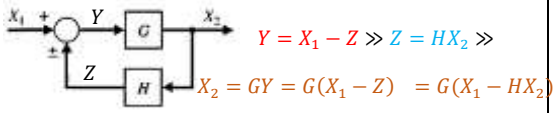
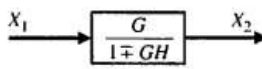
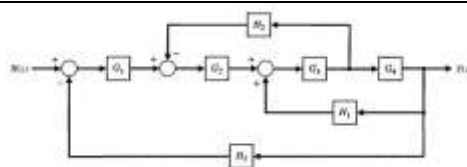


Block Diagram Reduction

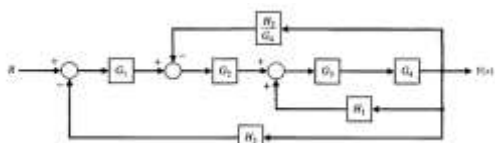
1. Blocks in series	 $X_2 = G_1 X_1 \ll X_3 = G_2 X_2$ $X_3 = G_1 G_2 X_1$	
2. Parallel blocks		
3. Moving a summing point ahead of a block	 $C(s) = \{R(s) \mp X(s)\}G(s)$	 $C(s) = \{R(s)G(s) \mp X(s)G(s)\}$
4. Moving a summing point behind of a block	 $C(s) = R(s)G(s) \mp X(s)$	 $C(s) = \left\{R(s) \mp X(s) \frac{1}{G(s)}\right\} G(s)$
5. Moving a pick off point ahead of a block		 $X_2 = G X_1 \ll X_1 = \frac{1}{G} X_2$
6. Moving a pick off point behind a block	 $X_2 = G X_1$	
7. Reducing a feedback loop	 $Y = X_1 - Z \gg Z = H X_2 \gg$ $X_2 = G Y = G (X_1 - Z) = G (X_1 - H X_2)$ $X_2 = \left(\frac{G}{1 \mp GH} \right) X_1$	

Example 1

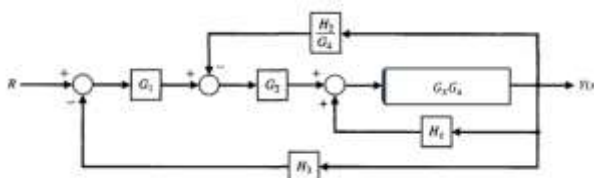
Reduce the given block diagram and find the overall transfer function of the system.



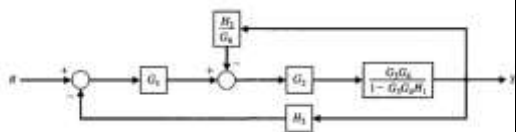
Moving the pickoff point ahead of the block G_4



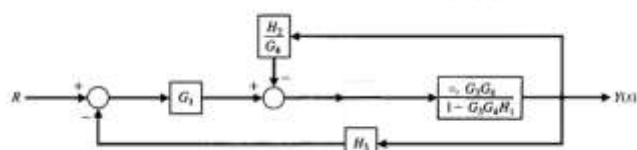
Combining the series blocks G_3 and G_4



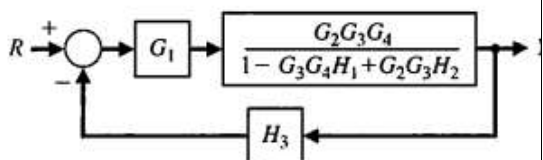
Eliminating the feedback loop



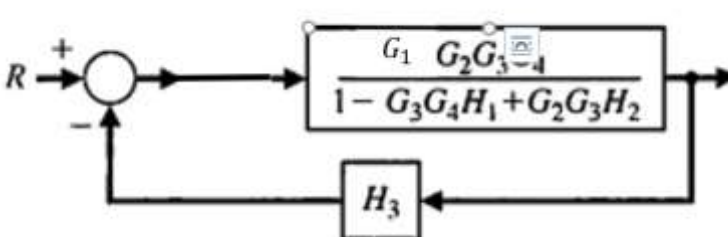
Combining the series blocks



Eliminating the feedback loop



Combining the series blocks



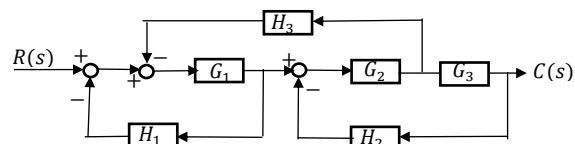
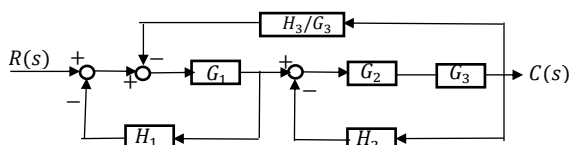
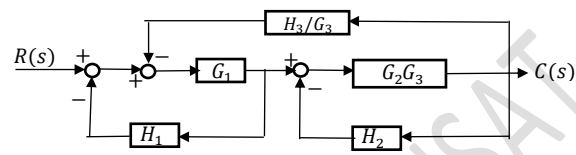
Eliminating the feedback loop

$$\frac{R(s)}{Y(s)} = \frac{G_1 G_2 G_3 G_4}{1 - G_3 G_4 H_2 + G_2 G_3 H_2 + G_1 G_2 G_3 G_4 H_3}$$

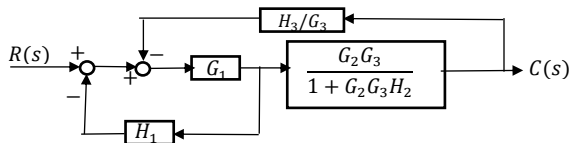
Example 2

Determine the closed loop transfer function

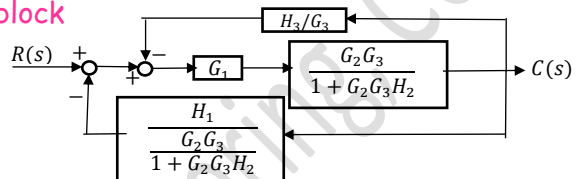
$$T(s) = \frac{C(s)}{R(s)}$$

Moving the pickoff point ahead of block G_3 Combining the series blocks G_2 and G_3 

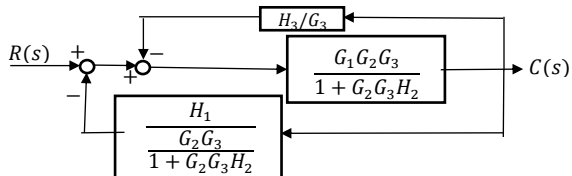
Eliminating the feedback loop



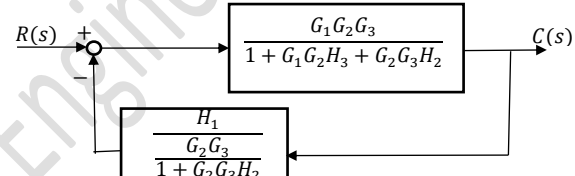
Moving the pickoff point ahead of the block



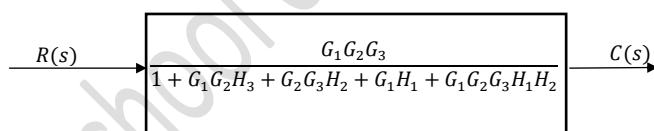
Combining the series block



Eliminating the feedback loop

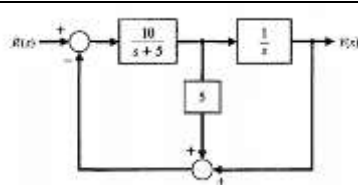


Eliminating the feed-back loop



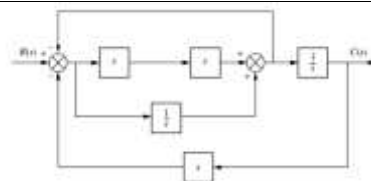
Exercise 1

Determine the closed loop transfer function $T(s) = \frac{Y(s)}{R(s)}$



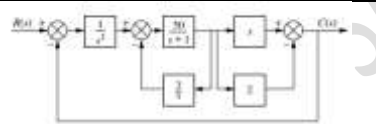
Exercise 2

Determine the closed loop transfer function $T(s) = \frac{C(s)}{R(s)}$



Exercise 3

Determine the closed loop transfer function $T(s) = \frac{C(s)}{R(s)}$



Exercise 4

Determine the closed loop transfer function $T(s) = \frac{C(s)}{R(s)}$

